# Legacy Cipher Migration Framework for Industrial Control Systems

A protocol-translation gateway from classical ciphers to AES-256-GCM for smart-grid telemetry, and a study of the vulnerabilities of the transition

[Student Name 1] ([Enrolment No. 1])  [Student Name 2] ([Enrolment No. 2])  [Student Name 3] ([Enrolment No. 3])

Guide: [Faculty Name] | [Course Code] - [Course Name] | School of Computer Science Engineering & Technology, [University Name]

# Agenda and presenters

01

## Problem and background

All members

02

## Legacy ciphers and how they fall

Member 1

03

## AES engine and secure envelope

Member 2

04

## Gateway, migration and evaluation

Member 3

05

## Results, demo and conclusion

All members

# Grid telemetry still runs on legacy links

- SCADA master polls RTUs in substations every few seconds
- RTUs report voltage, current, frequency, power and breaker status, and execute breaker commands
- Many links still use classical ciphers - or no encryption
- Devices cannot all be replaced at once: legacy and modern cryptography coexist for years

**15-30 years**

service life of field equipment

**1.2-19.2 kbaud**

typical legacy serial and radio links

**~225,000**

customers lost power in the 2015 Ukraine grid attack

# Problem statement and measurable objectives

*Build a protocol-translation gateway that wraps legacy, unencrypted or classically encrypted grid telemetry into AES-encrypted payloads - and identify, demonstrate and quantify the vulnerabilities of the transition, with measured mitigations.*

**O1** **Implement six legacy ciphers correctly**

100 % of textbook vectors; 0 round-trip failures on 10,000 frames per cipher

**O2** **Implement AES-128/192/256 and GCM from first principles**

100 % of FIPS-197 / GCM vectors; bit-identical to library on 500 cases

**O3** **Translate legacy telemetry to AES-GCM without data loss or delay**

0 corrupted readings in all 4 modes; median gateway latency < 1 ms

**O4** **Break every legacy cipher from captured traffic**

key recovered from <= 3 frames in < 1 s for each cipher

**O5** **Secure configuration blocks forged, replayed, tampered and downgraded frames**

0 % acceptance in every STRICT / PINNED / GCM scenario

**O6** **Identify and quantify transition vulnerabilities with mitigations**

>= 8 vulnerabilities, each with a measured mitigation

**O7** **Keep the secure envelope within legacy-link budgets**

overhead <= 64 B/frame; 8-device scan at 9600 baud < 2 s cycle

# Existing approaches and the research gap

| Approach | Works with legacy RTU? | Integrity | Limitation |
|---|---|---|---|
| Classical cipher (status quo) | n/a | none | broken from 1-2 frames |
| DNP3 Secure Authentication | no - firmware upgrade | HMAC | no confidentiality |
| IEC 62351 / Modbus-TCP Security | no - needs TLS stack | TLS | not for serial devices |
| AGA-12 / IEEE 1711.2, YASIR | yes - inline modules | yes | transition risks not quantified |
| LCMF (this project) | yes - translation gateway | AES-256-GCM + policy | last mile until RTU upgrade (TV1) |

**Gap: the risks of the migration itself - last mile, downgrade, replay laundering, AES misuse, nonce reuse, key provenance - are rarely measured.**

# Legacy ciphers and how they fall

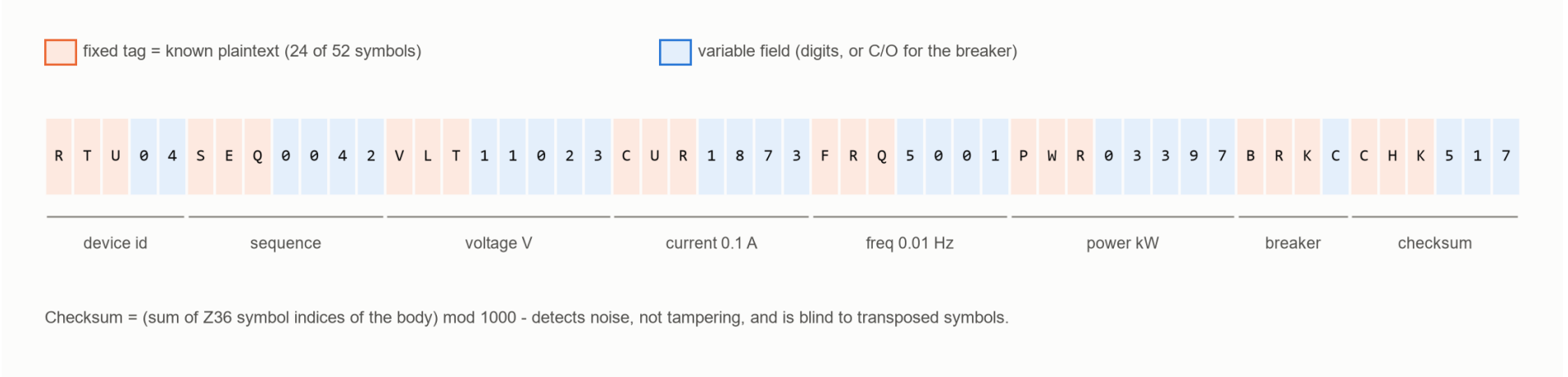
Member 1 - Legacy Cipher Suite, Cryptanalysis & Result Visualisation

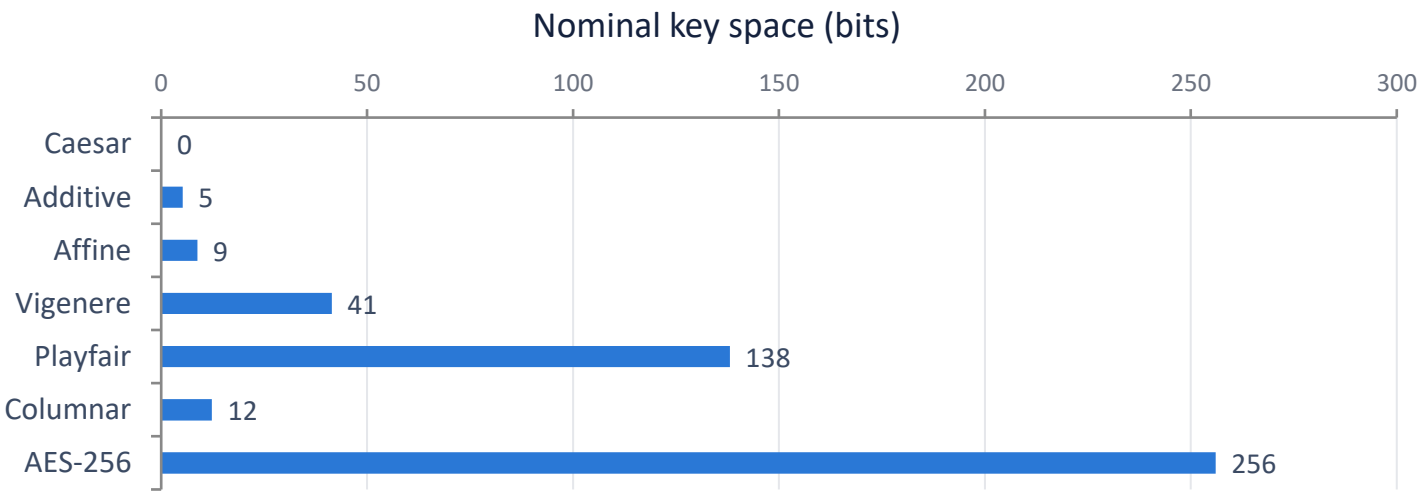
# Six legacy ciphers on one telemetry frame

| **Plaintext** | **RTU01SEQ1001VLT10791CUR1812FRQ5000PWR03217BRKCCHK000** |
|---|---|
| Caesar (k=3) | UWX34VHT4334YOW43AC4FXU4B45IUT8333SZU3654AEUNFFKN333 |
| Additive (k=17) | 8ABHI9V7IHHIC2AIHOQITB8IPIJW87MHHH6D8HKJIOS81TTY1HHH |
| Affine (a=5, b=8) | V5A4902Q9449F15943D9SAV989E7VQT444LKV4JE93NVWSSHW444 |
| Vigenere (key='GRIDLOCK' (L=8)) | XA23C6G07H846ZVB6OH4N8TBEIAI247A6HXZ2E5C7OJUVQERQH83 |
| Playfair (keyword='SUBSTATION' (6x6)) | VBBZYTNW2112WKD71673NBV0723EVR6Z3RYRXV1Y326TVJFRNJJ13R3R |
| Columnar (key='BREAKER' (cols=7)) | 007803CRQTURWB0U00150K0SV1201H119102CT11RQRR0ELCFP7K |

Key spaces over Z36: Caesar 1 - Additive 36 - Affine 432 - Columnar 7! = 5,040 - Vigenere 36^8 - Playfair 36! (about 2^138)

# Fixed frame grammar = free known plaintext




## <= 2 frames

and <= 85 ms to recover every legacy key - even Playfair's 2^138

# Cryptanalysis results: every key recovered

| Cipher | Attack | Frames | Keys tried | Time (ms) | Recovered key |
|---|---|---|---|---|---|
| Caesar | exhaustive key search | 1 | 36 | 0.6 | 3 |
| Additive | exhaustive key search | 1 | 36 | 0.6 | 17 |
| Affine | exhaustive key search | 1 | 432 | 7.5 | (5, 8) |
| Vigenere | template constraint propagation + checksum | 1 | 8 | 0.2 | GRIDLOCK |
| Playfair | keyword dictionary (178 words) | 1 | 178 | 5.0 | SUBSTATION |
| Columnar | exhaustive permutation search (L<=7) | 2 | 5,912 | 84.7 | [3, 0, 2, 5, 4, 1, 6] |

**56.3%**

of a new Playfair frame decrypted from ONE known frame (90.5% after 20) - no key needed

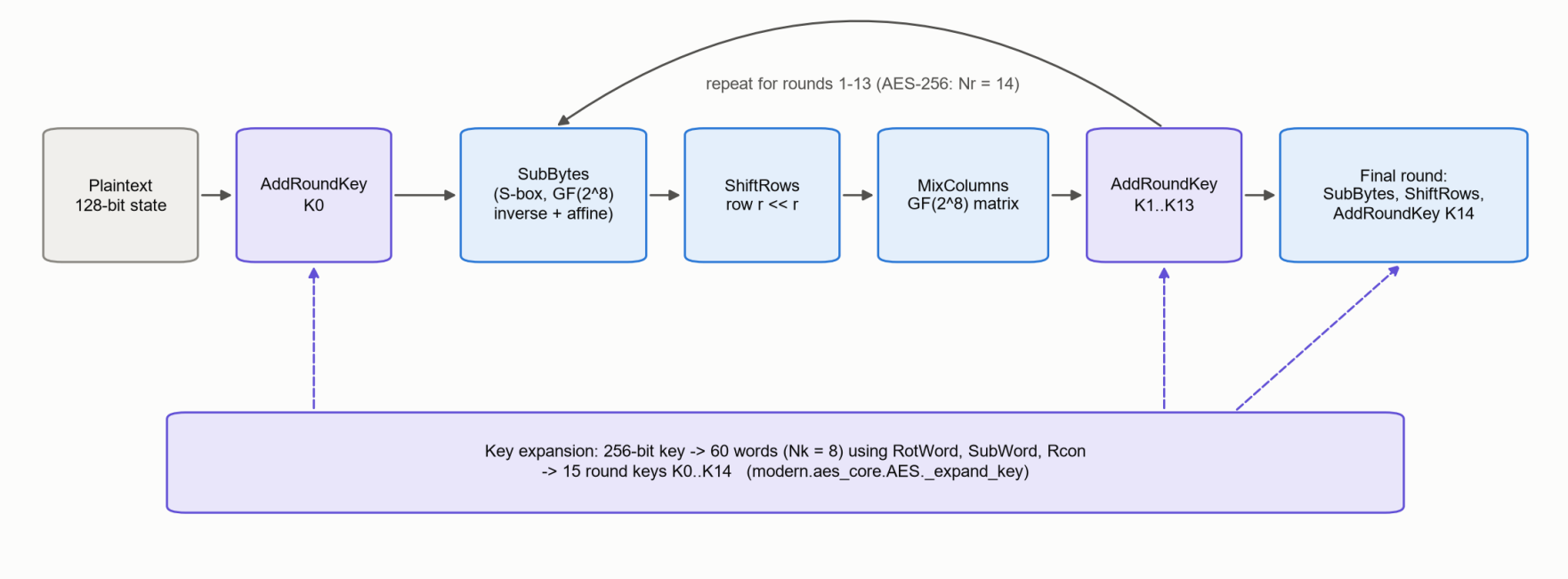
**Finding**

The legacy checksum is a plain sum: it cannot detect transposed digits, so a wrong Columnar key still gave checksum-valid frames.

# AES engine and secure envelope

Member 2 - Modern Cryptography Engine, Secure Envelope & Experiment Validation

# AES-256 and GCM from first principles




**15/15**

published test vectors pass (FIPS-197, GCM, textbook)

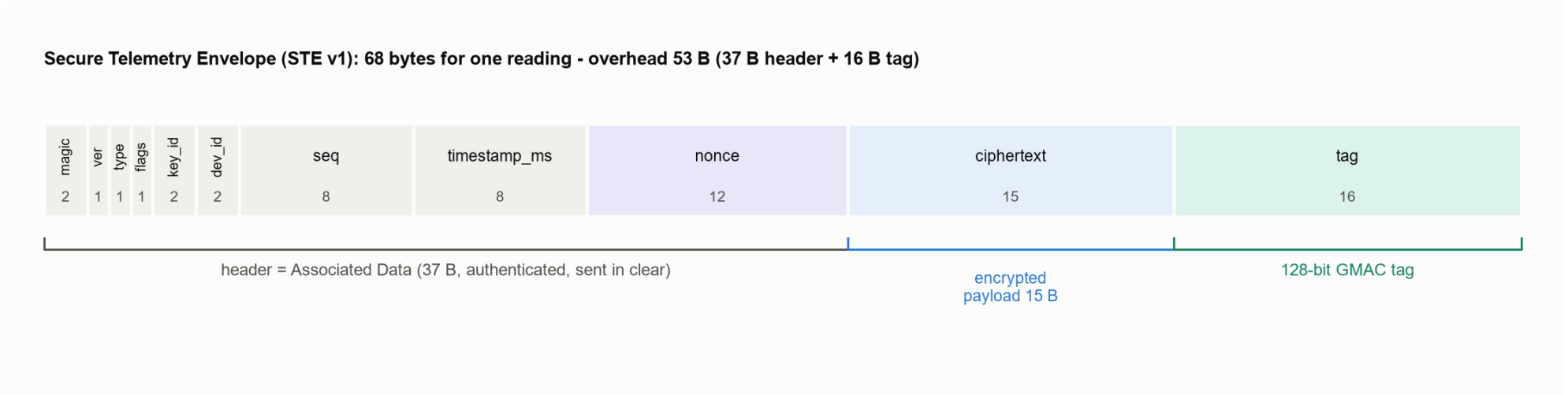
**500/500**

random cases bit-identical to the library

**70,000**

telemetry round trips, 0 failures

# Secure Telemetry Envelope: 68 bytes per reading




### 1 AAD

37-byte header authenticated by the GCM tag

### 2 Nonce

sender id || 64-bit counter, reserved and persisted before use

### 3 HKDF

one key per device, direction and generation from a random master

### 4 Replay

64-entry sliding window plus 30 s timestamp freshness

# Choosing AES is not enough (TV4-TV7)

TV4

## Malleable modes

**100% vs 0%**

blind breaker flip: AES-CTR without MAC vs AES-GCM

TV5

## ECB patterns

**3 ciphertexts**

for 1,440 status messages: breaker state inferred with 100% balanced accuracy (GCM: 50.0%)

TV6

## Nonce reuse

**630 reuses**

after reboots with RAM counters: 100% plaintext recovered; persistent counters: 0

TV7

## Legacy-derived keys

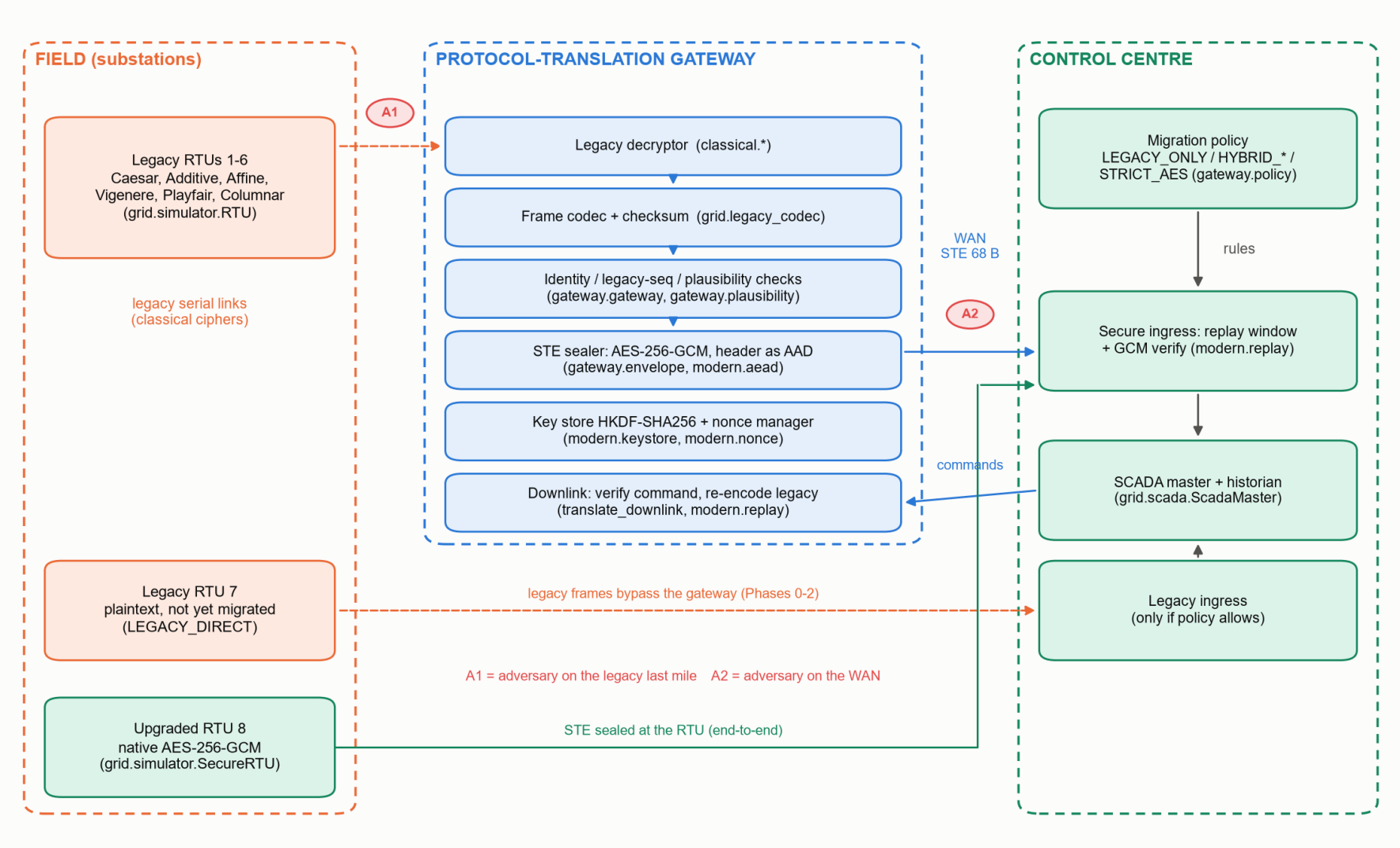
**101 guesses**

to find SHA-256("GRIDLOCK") from one envelope; HKDF random master: not found

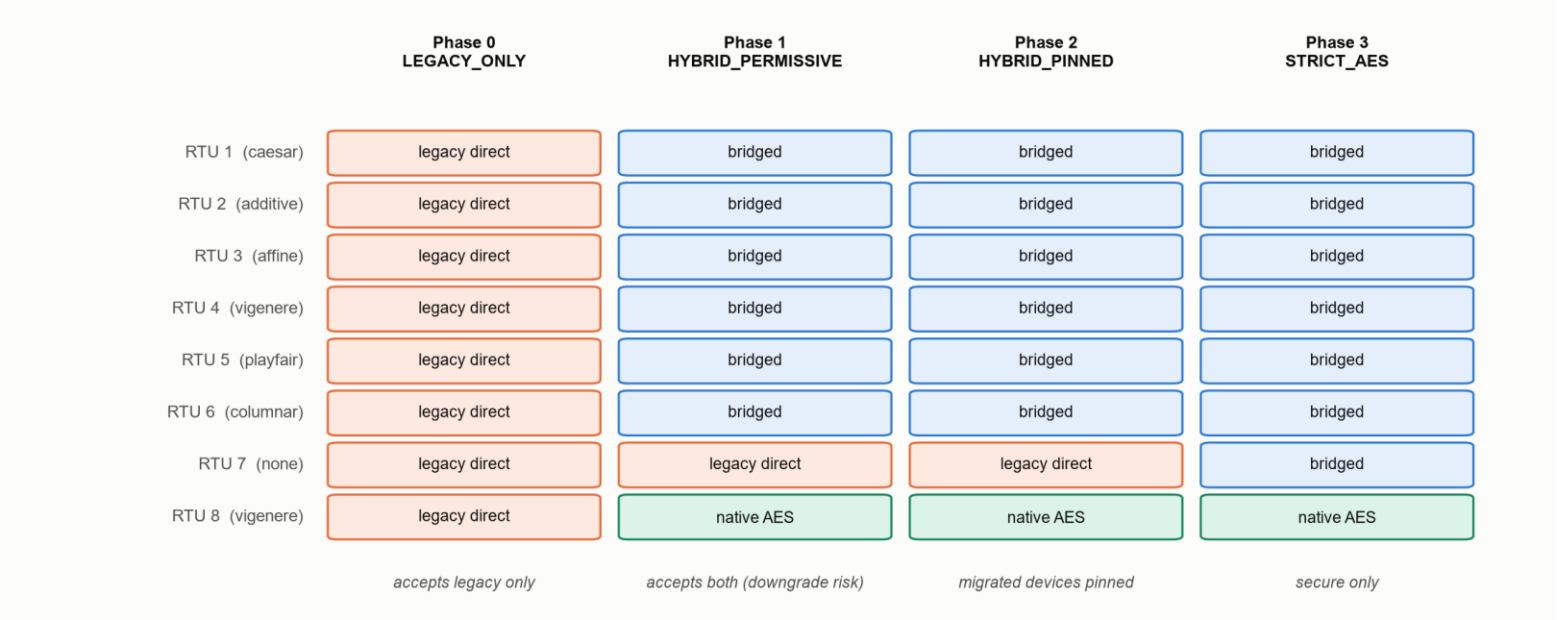
# Gateway, migration and evaluation

Member 3 - Smart-Grid Simulation, Gateway & System Evaluation

# Architecture of the migration gateway



# Four migration phases - and a downgrade trap




**100%**

forged legacy frames accepted in HYBRID_PERMISSIVE

**0%**

accepted once migrated devices are pinned (HYBRID_PINNED)

Legacy direct = talks legacy to SCADA | Bridged = legacy RTU behind the gateway | Native AES = upgraded RTU

# The legacy last mile: trust and freshness laundering

## TV1 - false data injected on the serial link

| Defence | Gross | Stealthy |
|---|---|---|
| Gateway only | 100% | 100% |
| Gateway + plausibility | 0% | 100% |
| Native AES at RTU | 0% | 0% |

Eavesdropper on the last mile already read 100% of the readings later sent inside AES envelopes.

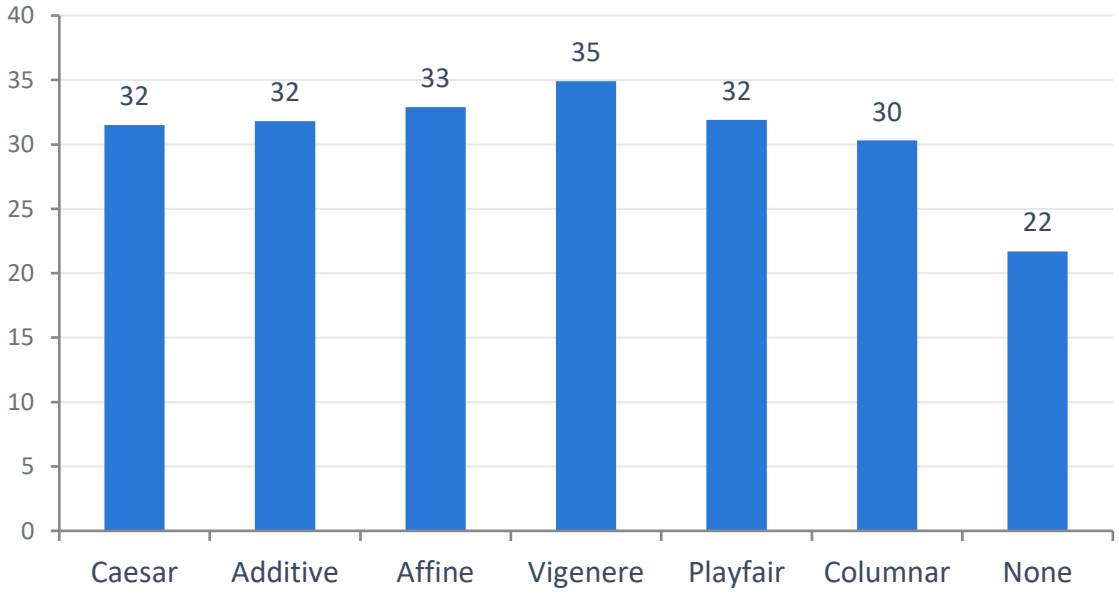
## TV3 - replay

| Scenario | Accepted |
|---|---|
| WAN replay of legacy frames - none (legacy) | 100% |
| WAN replay of secure envelopes - 64-bit seq + window + timestamp | 0% |
| Last-mile replay through the gateway - stateless gateway | 100% |
| Last-mile replay through the gateway - legacy sequence tracking at gateway | 0% |

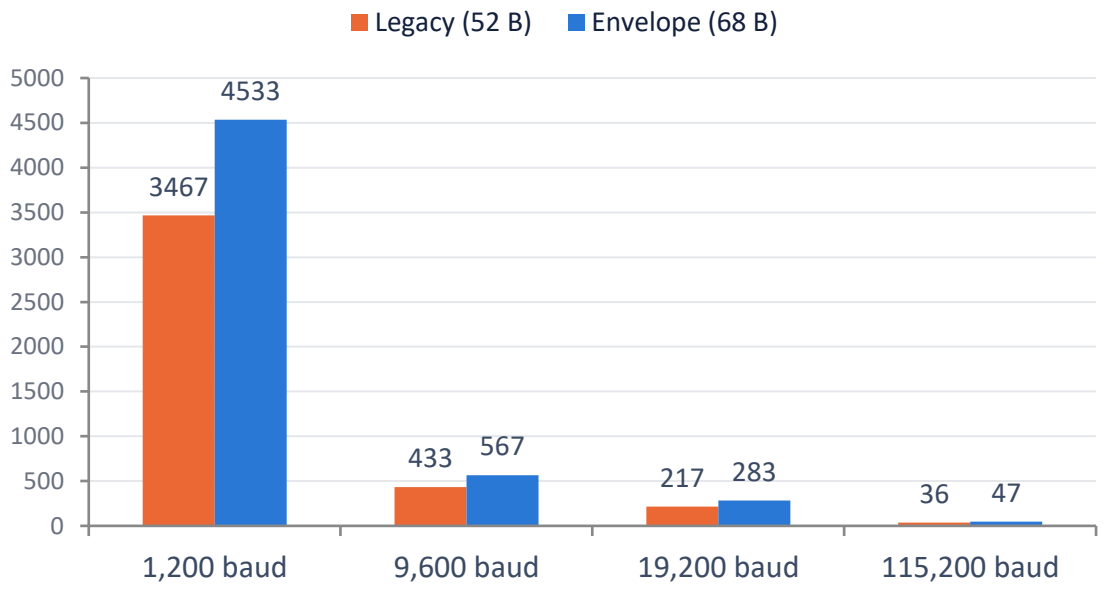
**A strong MAC can certify weak data: the gateway moves the trust boundary but does not remove the legacy link.**

# Correct and fast enough for real time

Gateway latency per frame, median (µs)




Time to poll 8 RTUs (ms); scan cycle 2,000 ms




**14,400**

frames delivered in 4 modes, 0 corrupted

**35 µs**

worst median latency (target < 1 ms)

**+53 B**

envelope overhead per reading (+30.8%)

# Every transition vulnerability, before and after mitigation



# Live demonstration

| | |
|---|---|
| `> python -m lcmf demo --interactive` | guided walk-through of all modules |
| `> python -m lcmf cipher affine encrypt RTU01SEQ0001 --a 5 --b 8` | any legacy cipher, any key |
| `> python -m lcmf attack` | every legacy key recovered |
| `> python -m lcmf simulate --mode STRICT_AES --polls 3` | gateway + SCADA in a migration phase |
| `> python -m lcmf experiments` | regenerates every number in the report |
| `> python -m pytest` | automated test suite |

# All seven objectives met

| ID | Objective | Measured result | Status |
|---|---|---|---|
| O1 | Implement six legacy ciphers correctly | 6/6 vectors; 0 failures in 70,000 round trips | **Met** |
| O2 | Implement AES-128/192/256 and GCM from first principles | 9/9 vectors; 500/500 identical | **Met** |
| O3 | Translate legacy telemetry to AES-GCM without data loss or delay | 0 corrupted of 14,400; worst median 34.9 us | **Met** |
| O4 | Break every legacy cipher from captured traffic | 6/6 keys; max 2 frames; max 85 ms | **Met** |
| O5 | Secure configuration blocks forged, replayed, tampered and downgraded frames | max acceptance 0.0 % across 6 scenarios | **Met** |
| O6 | Identify and quantify transition vulnerabilities with mitigations | 8 vulnerabilities (TV1-TV8) measured | **Met** |
| O7 | Keep the secure envelope within legacy-link budgets | 53 B overhead; scan 567 ms | **Met** |

# Conclusion: a migration playbook for utilities

1. Inventory legacy links, rank by exposure
2. Gateways next to RTUs, with plausibility and sequence tracking
3. Run HYBRID_PINNED; pin each device when migrated
4. Fresh random master keys + persistent nonce counters
5. Replace RTUs on the most exposed links first
6. Switch to STRICT_AES; delete legacy keys

## Future scope

Embedded gateway on real serial/radio links - Modbus RTU and DNP3 adapters with IEC 62351 on the control side - keys in an HSM/TPM - state-estimation or ML detection of stealthy false data - post-quantum key establishment

# Contribution of each team member

M1

## [Student Name 1]

[Enrolment No. 1]

### Legacy Cipher Suite, Cryptanalysis & Result Visualisation

- Designed the Z36 alphabet abstraction and the common ClassicalCipher interface
- Implemented Caesar, Additive, Affine, Vigenere, Playfair (5x5 and 6x6) and Columnar transposition
- Verified all ciphers against published textbook vectors and 70,000 telemetry round trips
- Built the cryptanalysis module: exhaustive search, template-constrained Vigenere attack, Playfair dictionary and codebook attacks, Friedman/Kasiski/chi-squared tools

1,269 lines of code | Slides 6-9 (legacy ciphers and how they fall)

M2

## [Student Name 2]

[Enrolment No. 2]

### Modern Cryptography Engine, Secure Envelope & Experiment Validation

- Implemented AES-128/192/256 from first principles (GF(2^8) arithmetic, derived S-box, key expansion)
- Implemented ECB, CBC, CTR and GCM (GHASH in GF(2^128)); cross-validated against the cryptography library
- Designed the 68-byte Secure Telemetry Envelope with header-as-AAD
- Implemented HKDF key store, persistent nonce manager and anti-replay window

1,184 lines of code | Slides 10-13 (AES engine, envelope, crypto pitfalls)

M3

## [Student Name 3]

[Enrolment No. 3]

### Smart-Grid Simulation, Gateway & System Evaluation

- Built the 11 kV feeder model, legacy RTUs, upgraded RTU, SCADA master and network simulator
- Implemented the bidirectional protocol-translation gateway and the four migration modes
- Implemented the plausibility filter and legacy-sequence tracking mitigations
- Designed and ran system transition experiments TV1-TV3 and TV8

1,321 lines of code | Slides 14-18 and 20 (gateway, migration, transition vulnerabilities, live demo)

# Thank you - questions?

Code, results, report and slides are generated from one repository:  python -m lcmf demo

Key references: NIST SP 800-82 Rev. 3  -  NISTIR 7628  -  FIPS 197  -  NIST SP 800-38D  -  IEEE 1711.2  -  IEC 62351  -  Lee et al., Ukraine grid attack analysis (2016)